



Figure 1: Row-based vs column based databases



Figure 2: MonetDB installation with Python

So, the choice of the database will depend upon the features that are required. Column based databases are preferred for statistical operations, where we have a huge amount of data to manage and manipulate.

MonetDB

MonetDB is a free, open source database management system developed by the CGI database research group. The main use case of this database is for data warehousing, where there is a huge amount of data to manage. It uses a vertical fragmentation and columnar execution based engine, and is mainly used for Big Data science. MonetDB + R language integration is popular among data scientists.

Installation of MonetDB in Python

There are different ways to install MonetDB. We can do a standalone installation or install the thin clients provided in different languages like Python and R. Here, we will explore how to use Python for the installation of MonetDB, just for demo purposes.

1. Install the Python-Monetdb package; we are using the Pip utility.
2. Run the command given below in a CLI prompt:

```
pip install monetdblite
```

3. We can also install standalone MonetDB in Windows. Download the .msi file from <https://www.monetdb.org/downloads/Windows/Jul2017-SP3/>.
4. Figure 3 gives the snapshot for Windows installation.
5. To connect to a database, use the code given below in Python:

```
# create a new database or connect to an existing database
in /tmp/db
import monetdblite
conn = monetdblite.connect('/tmp/db')
```



Figure 3: MonetDB Windows installation

	Microsoft SQL Server	MonetDB
Details	it's Microsoft relational database which store data in row format	it's an open source column-based database system
Database model	Relational DBMS	Relational DBMS
Implementation	C++	C
OS	Linux, Windows	FreeBSD, Linux, OS, Solaris, Windows
Transaction concepts	ACID	ACID
Concurrency	Yes	Yes
User concepts	Fine grained access rights based on SQL-Standard	Fine grained access rights based on SQL-Standard
Language Support	All Major languages including Visual Basic	All Major languages including Visual Basic
License	commercial	Open Source

Figure 4: A comparison between a traditional database and MonetDB

To create a cursor to access tables and databases, use the following code:

```
# create a new cursor
c = conn.cursor()

# query the database
c.execute('SELECT * FROM tables')
# fetch the results and print them
print(c.fetchall())
```

Likewise, we can explore more commands from the MonetDB Command Reference Guide. 

References

- [1] <<https://db-engines.com/en/system/Microsoft+SQL+Server%3BMonetDB->>
- [2] <https://www.monetdb.org/wiki/MonetDB:Getting_started>
- [3] <<https://www.monetdb.org/Documentation/UserGuide/MonetDB-R>>

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